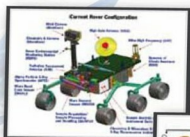


MODEL DYNAMIC SYSTEMS

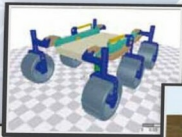


Simulate in real time

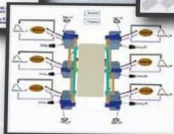
ELECTRICAL | MECHANICAL | THERMAL | HYDRAULICS



Conceptual Model



3D Animation



3D Mechanics



Dynamic Model using 20-sim editor



Prototype Model using 20-sim 4C

from concept to PROTOTYPE

ExoMars (Exobiology on Mars) is a space mission under development at the European Space Agency (ESA), which will send a robotic rover to the surface of Mars. The rover will carry a comprehensive suite of analytical instruments dedicated to exobiology research. The rover will make large use of electric actuators and related motion control systems.

A Rover model was created using the 3D Mechanics toolbox and was exported to the 20-sim Editor with all properties of mass, moment of inertia and all joints required for the special ESA model. A hierarchical approach was followed using bond graph, block diagram and equations in a port based modelling environment. Control and actuators inputs were given to the dynamic model from Mechatronics toolbox. The model was used to test various types of locomotion.

To support testing of new motion controllers, a generic test bench has been developed at ESTEC. The purpose of the test bench is to test the proper working of a motion control unit (MCU), and then characterize the most important performance parameters. To measure the performance of the MCU and motors, a data acquisition system was implemented based on a TS-7300 board and the rapid prototyping environment 20-sim 4C.

Many options were investigated at ESTEC regarding an integrated software and hardware solution from the contemporaries. Finally 20-sim and 4C were chosen as being adequately fulfilling the technical needs and being dramatically lower in cost with respect to the alternative solutions.

...realize your **MECHATRONICS** applications with **20-sim**
Get your evaluation copy today from www.20sim.com

Authorized distributor of 20-sim for Asia Pacific Region



ASIC Design



Dynamic Simulation



EM Simulation



Rapid Prototyping



System Prototyping



EMF Safety